SEFS LAB HPC COMMAND CHEAT SHEET

General command guide for NC State HPC (Hazel) using Slurm scheduler

Cluster access: `ssh <unityid>@login.hpc.ncsu.edu` | **OS:** RHEL 9.2

Author: Michelle Pretorius | **Updated:** Aug 2026 | **Scheduler:** Slurm

1. FILESYSTEMS & STORAGE DIRECTORIES

Location	Path & Quota	Purpose / Policy
Home	`~/` or `/home/<unityid>` 1 GB QUOTA	Scripts, small apps. Backed up daily.
Scratch	`/share/doserlab/<unityid>` 20 TB QUOTA	Job execution, raw results. **Not backed up!** Purged after 30 days inactive.
App dir	`/usr/local/usrapps/doserlab/jwdoser`	SEFS Lab shared software and R packages repository.
Research	`/rs1` or `/rsstu` 2TB - 30TB	Long-term project data storage across nodes.

2. CONNECTION & INTERACTIVE SFTP

Command	Description
`ssh <unityid>@login.hpc.ncsu.edu`	Log in to Hazel via SSH (requires Duo 2FA).
`sftp <unityid>@login.hpc.ncsu.edu`	Open interactive secure file transfer session.
`exit` / `Ctrl+D`	Terminate SSH or SFTP connection.
`hostname`	Identify current node (`loginXX` vs compute node).
`whoami` / `clear`	Print active UnityID / Clear terminal screen.

Interactive SFTP sub-commands (inside `sftp>` prompt):

Command	Description
`put <local> <remote>`	Upload file to HPC (add `-r` for folders).
`get <remote> <local>`	Download file from HPC (add `-r` for folders).
`lcd <path>` / `lpwd`	Change/print working directory on **local machine**.

3. DIRECTORY & FILE OPERATIONS

Command	Description / Key Options
`pwd`	Print current full working directory path.
`cd <path>`	Change directory (`cd ~` home, `cd ..` parent).
`ls -ltr`	List files in long format sorted by modification time.
`mkdir <folder>`	Create a new directory (`rmdir` removes empty dir).
`nano <file>`	Command-line text editor (`Ctrl+X` to exit/save).
`cat <file>`	Print entire file contents to terminal screen.
`head -n 20 <f>`	Print first 20 lines of a file.
`tail -f <f>`	Print last 10 lines of a file (`-f` monitors live).
`less <file>`	Page through file line-by-line (press `q` to exit).
`grep “str” <f>`	Search pattern inside file (e.g. `ls | grep txt`).
`cp -r <s> <d>`	Copy file or directory (`-r` recursive).
`mv <s> <d>`	Move or rename a file or directory.
`rm <file>`	Permanently delete file (**Warning: cannot undo!**).

4. QUOTA & MODULE MANAGEMENT

Command	Description
`quota -s`	Display home directory quota and space used.
`quota_display`	NC State utility to check group scratch quota on `/share` .
`du -sh .`	Summarize total space used by current directory.
`module avail`	List all software modules available on Hazel.
`module load R`	Load specific module into active session (e.g. R, gcc).
`module list`	Show software modules currently loaded.
`module purge`	Unload **all** modules (best practice at start of scripts).

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5. SLURM BATCH SCRIPT DIRECTIVES (#SBATCH)

Directive Flag	Description / Example
-job-name=<name>	Name displayed in queue (e.g. myjob).
-output=stdout.%j	Standard output log file (%j = JOBID).
-error=stderr.%j	Standard error log file.
-ntasks=1	Number of tasks/processes (MPI ranks).
-cpus-per-task=4	Number of CPU cores per task (threads).
-nodes=1	Number of compute nodes requested.
-time=04:00:00	Wall clock limit (HH:MM:SS or D-HH:MM:SS).
-mem=16G	Memory per node (e.g., 16G or 16000M).
-partition=compute	Partition selection: compute (CPU) or gpu .
-qos=normal	QOS tier: normal (max 4 days) or long (max 10 days).
-array=1-10	Submit job array with task indices 1 through 10.

6. TEMPLATE SLURM BATCH SCRIPT

```
#!/bin/bash
#SBATCH --job-name=myjob
#SBATCH --output=stdout.%j
#SBATCH --error=stderr.%j
#SBATCH --ntasks=1
#SBATCH --cpus-per-task=4
#SBATCH --time=02:00:00
#SBATCH --mem=16G
#SBATCH --partition=compute
#SBATCH --qos=normal

# Explicitly switch to submission directory
cd $SLURM_SUBMIT_DIR

# Clean environment & load modules
module purge
module load R

# Execute R script
Rscript ~/scripts/my_model.R
```

7. SLURM JOB MANAGEMENT COMMANDS

Command	Description
sbatch submit.sh	Submit batch script to Slurm execution queue.
salloc	Start an interactive session on a compute node.
srun <exec>	Launch parallel job step (MPI) or command.
squeue	View status (ST) of your queued/running jobs.
sjs <JOBID>	View live CPU %, RAM, and disk I/O for running job.
scancel <JOBID>	Cancel running job (scancel -u $USER cancels all).
sacct -j <JOBID>	View accounting details/exit state for completed job.
seff <JOBID>	Display CPU and memory efficiency utilization report.
sa	Show user account allocations and allowed QOS.
sqos	View QOS policies, priorities, and max wall time.
si	Show compute node availability across partitions.

8. SLURM STATES & ENVIRONMENT VARIABLES

Code	State	Description
PD	Pending	Waiting for requested cluster resources or QOS slot.
R	Running	Job is actively executing on allocated compute node(s).
CD	Completed	Job finished successfully with exit code 0.
F / TO	Failed / Timeout	Non-zero exit code or exceeded requested -time limit.
OOM	Out of memory	Job process was killed for exceeding -mem limit.

Slurm environment variables (used inside scripts):

$SLURM_SUBMIT_DIR	Directory path where sbatch was executed.
$SLURM_ARRAY_TASK_ID	Current task index for Job Array (e.g. 1, 2, 3...).
$SLURM_JOB_ID	Unique numeric ID assigned to the job.